LL(1) Parser Assignment Report

Course: COMP-451: Compiler Construction
Assignment: LL(1) Parser Implementation in C
Student: ____________________

Part 1: Grammar Transformation

E → T E'
E' → + T E' | ε
T → F T'
T' → * F T' | ε
F → (E) | id

Part 2: FIRST & FOLLOW Sets

FIRST(E) = { (, id }
FIRST(E') = { +, ε }
FIRST(T) = { (, id }
FIRST(T') = { *, ε }
FIRST(F) = { (, id }

FOLLOW(E) = {), $ }
FOLLOW(E') = {), $ }
FOLLOW(T) = { +,), $ }
FOLLOW(T') = { +,), $ }
FOLLOW(F) = { *, +,), $ }

Part 3: LL(1) Parsing Table

NT	id	+	*	(	)	$
E	E→TE'			E→TE'		
E'		E'→+TE'			E'→ε	E'→ε
T	T→FT'			T→FT'		
T'		T'→ε	T'→*FT'		T'→ε	T'→ε
F	F→id			F→(E)		

Part 4: C Implementation

```
#include
#include
#include

char input[100];
int i = 0;

void E(); void E1(); void T(); void T1(); void F();

void match(char c){
if(input[i]==c) i++;
else { printf("Rejected\n"); exit(0); }
}

void F(){
if(input[i]=='i'){
printf("F->id\n");
match('i');
} else if(input[i]=='('){
printf("F->(E)\n");
match('(');
E();
match(')');
} else { exit(0); }
}

void T1(){
if(input[i]=='*'){
printf("T'->*FT'\n");
match('*');
F();
T1();
} else printf("T'->ε\n");
}

void T(){
printf("T->FT'\n");
F();
T1();
}

void E1(){
if(input[i]=='+'){
printf("E'->+TE'\n");
match('+');
T();
E1();
} else printf("E'->ε\n");
}

void E(){
printf("E->TE'\n");
T();
E1();
```

```
}

int main(){
scanf("%s", input);
strcat(input, "$");

printf("Parsing Started\n");
E();

if(input[i]=='$') printf("Accepted\n");
else printf("Rejected\n");
}
```